Communication Module – Technical Documentation

Project: TaskFlow – Intelligent Task Allocation & Tracking System
Module: Communication · **Child Modules:** Notifications (incl. deadline reminders engine), Mail
Version: 1.0 · **Last Updated:** 31-Oct-2026
Backend: ASP.NET Core Web API (.NET 10) · **Frontend:** Angular · **Database:** SQLite (EF Core)
Author: Jojo George (jojogeorge3344@gmail.com)
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✅ Target framework confirmed: `net10.0` (.NET 10), from `backend/src/TaskFlow.Api/TaskFlow.Api.csproj`.

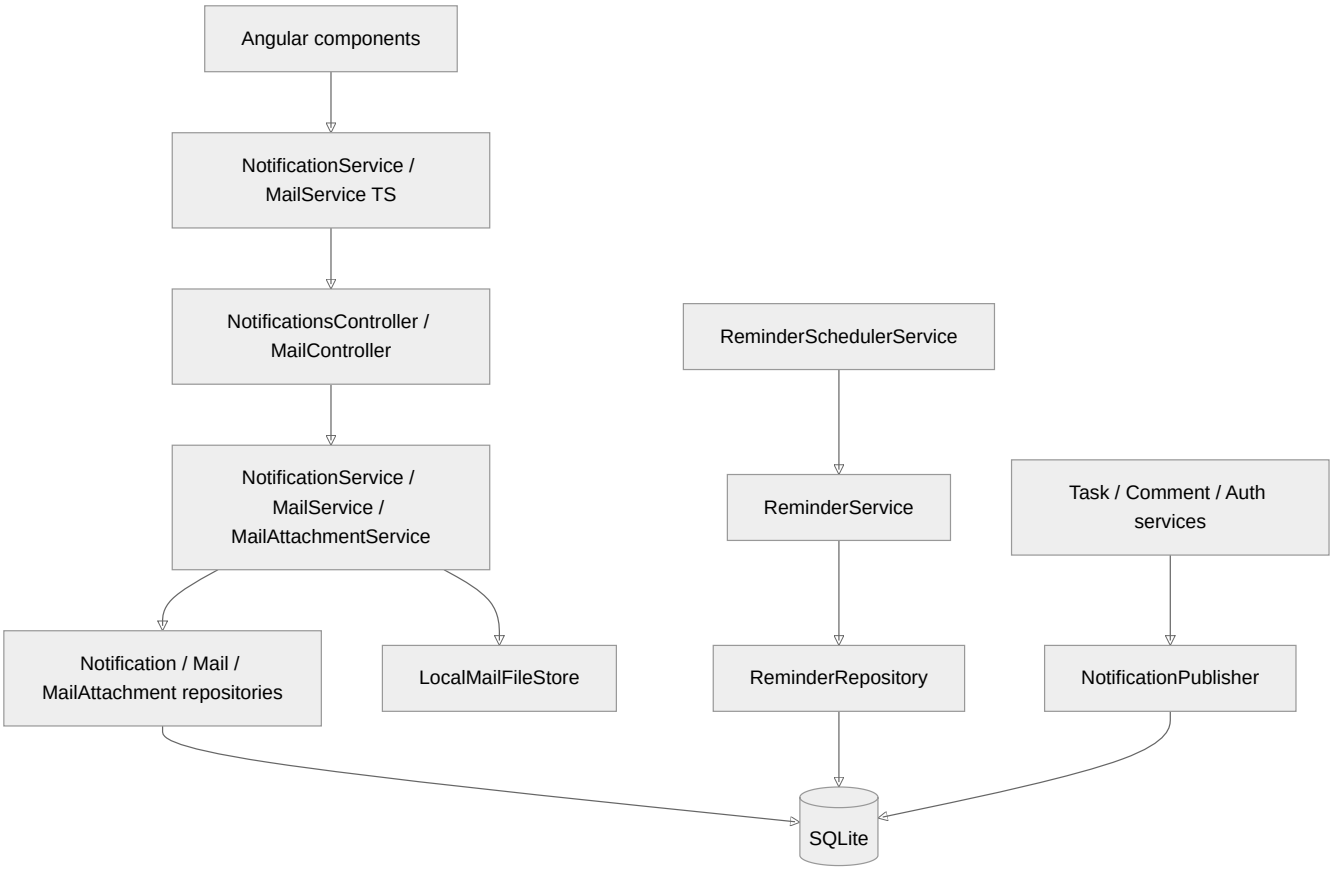
1. Technical Overview

The Communication module has these parts, split over two child modules:

- **Notifications (read side)** – child 05.1: `NotificationsController` → `NotificationService` → `NotificationRepository`. It serves the caller's own rows from the `Notifications` table for the Notification Center (BRD Screen 12).
- **Notification publishing (write side)** – child 05.1: `INotificationPublisher` / `NotificationPublisher` is an append-only seam that other modules call (Monitoring, Auth). It skips writes while the `Notification.InAppEnabled` application setting is `false`.
- **Deadline reminder engine** – child 05.1: `ReminderSchedulerService` is a `BackgroundService` running a `PeriodicTimer`. On each tick it calls `ReminderService.GenerateDueAsync()`, which records each due reminder once in `ReminderHistory` and then writes a `Notification` and, optionally, a system `MailMessage`.
- **Internal mail** – child 05.2: `MailController` → `MailService` / `MailAttachmentService` → `MailRepository` / `MailAttachmentRepository`. Each user has their own copy of a mail (`MailboxItem`), so read state, flags and trash are personal. Attachment bytes are stored on disk by `LocalMailFileStore`.

Every endpoint is under `/api/v1/…`, requires a JWT (`[Authorize]`) and a permission policy `perm:<key>`, and resolves the caller from the `ClaimTypes.NameIdentifier` claim.

2. Architecture Flow



3. Backend File Structure

Module-level summary. The full file lists are in each child document.

Layer	Path	Child
Controller	`TaskFlow.Api/Controllers/Communication/`	`NotificationsController.cs` (05.1), `MailController.cs` (05.2)
Services	`TaskFlow.Api/Services/Communication/Notification/` (`Notifications/`, `Publishing/`, `Reminders/`)	05.1
Services	`TaskFlow.Api/Services/Communication/Mail/` (`Messages/`, `Attachments/`)	05.2

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Layer	Path	Child
Repositories	`TaskFlow.Api/Repositories/Communication/Notification/` (`Notifications/`, `Reminders/`)	05.1
Repositories	`TaskFlow.Api/Repositories/Communication/Mail/` (`Messages/`, `Attachments/`)	05.2
DTOs	`TaskFlow.Api/Dtos/Communication/Notification/`, `TaskFlow.Api/Dtos/Communication/Mail/` (+ `Validators/`)	05.1, 05.2
Rules	`TaskFlow.Api/Common/Rules/`	`ReminderCalculator.cs` (05.1); `MailRules.cs`, `EmailPolicy.cs` (05.2)
Exceptions	`TaskFlow.Api/Common/Exceptions/`	`MailValidationException.cs` (05.2); `AppException.cs` (`ConflictException`, shared)
Entities / Enums	`TaskFlow.Api/Models/Entities/`, `TaskFlow.Api/Models/Enums/`	Listed per child
Integration	`TaskFlow.Api/Integrations/Storage/`	`IMailFileStore.cs`, `LocalMailFileStore.cs` (05.2)
Config	`TaskFlow.Api/Config/`	`ReminderEngineOptions.cs` (05.1); `MailOptions.cs` (05.1 and 05.2); `ServiceRegistration.cs`, `IntegrationRegistration.cs` (shared)
Authorization	`TaskFlow.Api/Common/Security/`	`PermissionKeys.cs`, `PermissionCatalog.cs`, `PermissionPolicy.cs` (prefix `perm:`) – shared
Authorization (audit)	`TaskFlow.Api/Authorization/Auditing/`	`DeniedActionAuditHandler.cs` (logs every 403) – shared
Persistence	`TaskFlow.Api/Persistence/`	`AppDbContext.cs` (shared); migrations `20261013075057_AddTaskTracking` (creates `Notifications`), `20261017062735_AddReminderEngine` (creates `ReminderHistory`, `ApplicationSettings`), `20261017074817_AddEmailDelivery` (creates `EmailDeliveryLogs`), `20261017090431_AddInternalMail` (creates the four Mail tables, drops `EmailDeliveryLogs`)
Cross-cutting	`TaskFlow.Api/Middleware/`, `TaskFlow.Api/Filters/`, `TaskFlow.Api/Common/Models/`	`GlobalExceptionHandler.cs`, `ValidationFilter.cs`, `PagedResult.cs` – shared

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4. Frontend File Structure

Module-level summary. The component, service and model files are listed in each child document.

Type	Path	Purpose
Routes	`src/app/modules/communication/communication.routes.ts`	`/mail` (requires `mail.read`), `/notifications` (requires `notifications.view`), both lazy-loaded with `permissionGuard`
Child folder	`src/app/modules/communication/notification/`	Notification Center (05.1)
Child folder	`src/app/modules/communication/mail/`	Mail screen and compose panel (05.2)
Route / Guard	`src/app/shared/auth/auth.guard.ts` (`permissionGuard`)	If the user lacks the key, redirects to their dashboard
Permission constants	`src/app/shared/auth/permissions.ts`	`MailRead`, `MailCompose`, `MailOrganize`, `MailAttachments`, `NotificationsView`, `NotificationsManage`
Menu	`src/app/layouts/nav-items.ts`	"Communication Management" group: Mail (`mail.read`, badge `mail`), Notifications (`notifications.view`)
Shell	`src/app/layouts/shell.component.ts/.html`	Header bell (`canSeeNotifications`) with unread dot; Mail badge. Counts refresh on navigation and on a 15 s interval (`SESSION_CHECK_INTERVAL_MS`).

5. Child Modules

#	Child Module	Main endpoints (count)	Key permissions	Document
5.1	Notifications (incl. deadline reminders engine)	`GET /api/v1/notifications`, `GET …/unread-count`, `POST …/{id}/read`, `POST …/read-all` (4). The reminder engine has no endpoint.	`notifications.view`, `notifications.manage`	`05.1-Notifications.md`
5.2	Mail	`GET /api/v1/mail`, `/summary`, `/unread-count`, `/items/{id}` (open, flags, trash, restore, delete, compose), `/send`, `/drafts`, `/contacts`, `/attachments` (16)	`mail.read`, `mail.compose`, `mail.organize`, `mail.attachments`	`05.2-Mail.md`

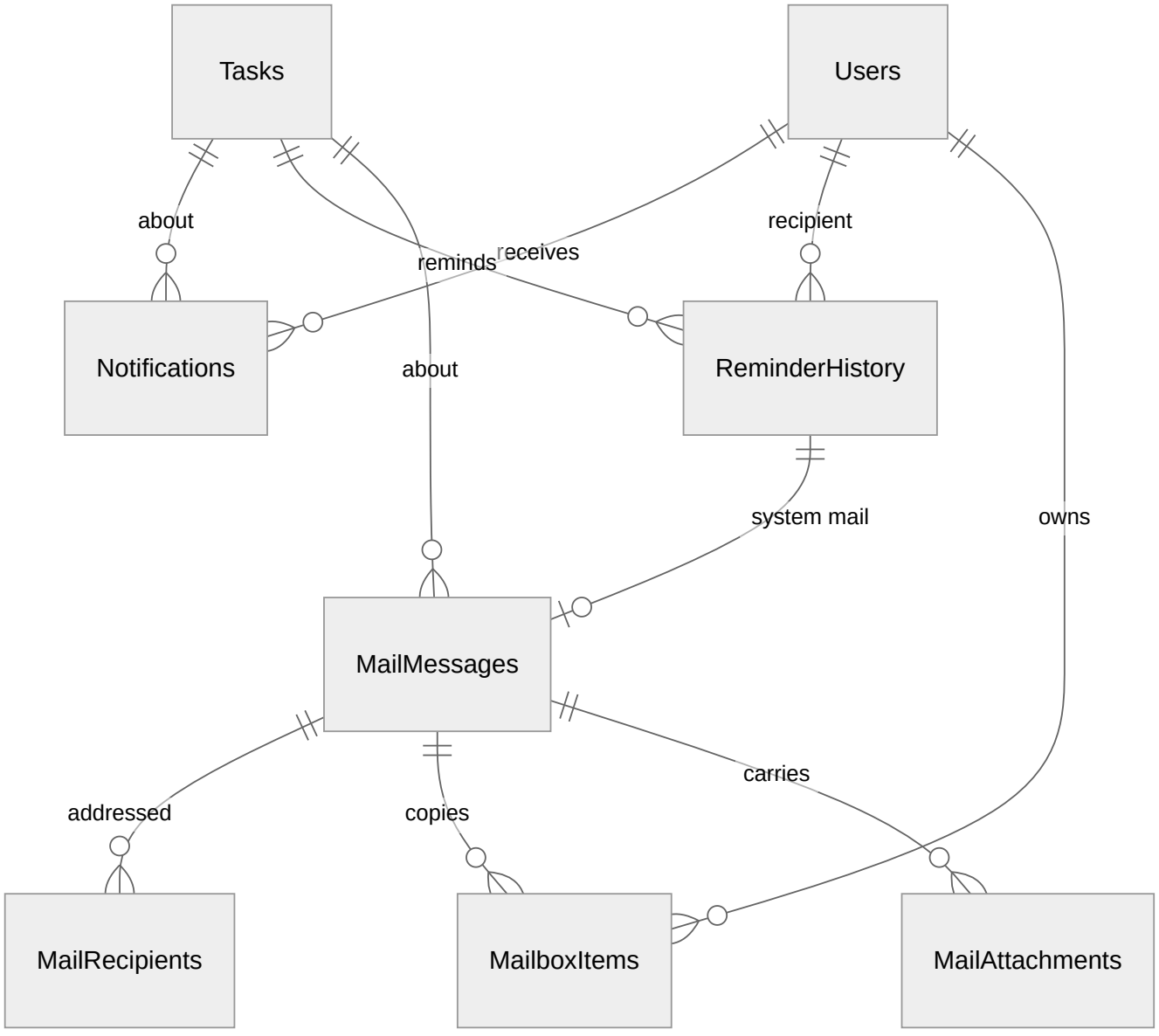
The child documents are in the `05-Communication/` folder next to this file.

6. Database Design

Column types are given as C# type → SQLite storage. Enums that use `HasConversion<string>()` are stored as TEXT. Column-level detail is in each child document.

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Table	Owning child	Purpose
`Notifications`	05.1 Notifications	One in-app alert per user
`ReminderHistory`	05.1 Notifications	One row per delivered reminder; unique key prevents duplicates
`MailMessages`	05.2 Mail	Mail content, conversation, draft fields; system mails link to `ReminderHistory`
`MailRecipients`	05.2 Mail	To / Cc of each message
`MailboxItems`	05.2 Mail	Each user's personal copy of a message (folder, flags, trash)
`MailAttachments`	05.2 Mail	File metadata; bytes are on disk



Cross-child relationship: `MailMessages.ReminderHistoryId → ReminderHistory.Id` (FK, Restrict, unique filtered index) – one system mail per reminder. The reminder engine (05.1) writes `MailMessages`, `MailRecipients` and `MailboxItems` rows owned by Mail (05.2).

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Shared reads (owned by other modules): `Tasks` (`Status`, `Deadline`, `FirstReminderDays`, `SecondReminderDays`, `OverdueReminderFrequency`, `CreatedByUserId`, `IsDeleted`) and `TaskRecipients` (Monitoring module); `Users`; `ApplicationSettings` keys `Reminder.*`, `Notification.MailEnabled`, `Notification.InAppEnabled`, `Application.Timezone` (Settings module).

7. Security & Authorization

- **Authentication:** JWT bearer token plus refresh-token cookie. The class-level `[Authorize]` on both controllers requires a signed-in user. If the `NameIdentifier` claim is not an integer, the endpoint returns `401`.
- **Authorization:** each action carries `[Authorize(Policy = PermissionPolicy.Prefix + PermissionKeys.X)]`, which resolves to the `perm:<key>` policy. The keys and the role defaults are defined in `PermissionCatalog.cs` (section "Communication Management"). Refusals are logged by `DeniedActionAuditHandler` ("Denied action: user … attempted …").
- **Data scope:** always **Own**. A record belonging to someone else returns **404, never 403**. The one exception is `GET /mail/contacts`, which searches the whole directory of active users (see 05.2).
- **Frontend:** `permissionGuard` on `/mail` and `/notifications`. Menu entries and the header bell are built from `has(Permissions.X)`. Components hide buttons with computed signals from `AuthService.hasPermission`.

Permission Key	Child	Detail
`notifications.view`	05.1	Key → endpoints → UI elements in `05.1-Notifications.md` §7
`notifications.manage`	05.1	Same
`mail.read`	05.2	Key → endpoints → UI elements in `05.2-Mail.md` §7
`mail.compose`	05.2	Same
`mail.organize`	05.2	Same
`mail.attachments`	05.2	Same

8. Validations & Error Handling

Shared rules that apply to every endpoint in the module. Child-specific rules are in each child document.

Rule	Layer	HTTP Code	Error Message
Not signed in, or bad subject claim	`[Authorize]` / controller	401	–
No permission	Policy `perm:<key>`	403	Forbidden (empty body; logged by `DeniedActionAuditHandler`)
Notification, mail item, draft, parent or attachment not owned or not found	Service / Controller	404	– (empty `NotFound()`)

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Rule	Layer	HTTP Code	Error Message
Concurrent update of the same row	`GlobalExceptionHandler` (`DbUpdateConcurrencyException`)	409	Someone else changed this record while you were saving. Reload it and try again.
Unhandled error	`GlobalExceptionHandler`	500	Something went wrong. Please try again.

Errors are returned by `GlobalExceptionHandler` (`Middleware/GlobalExceptionHandler.cs`) as RFC 7807 problem details. `ValidationFilter` runs the FluentValidation validators before each action.

9. Integrations / Side Effects

- **Between the children:** the reminder engine (05.1) and the comment service write system mails (`IsSystem = true`) into Mail (05.2) when the `Notification.MailEnabled` setting ("Also deliver reminders to Mail") is on.
- **Other modules → Notifications:** Monitoring (`TaskService`, `TaskCommentService`) and Auth (`MfaService`, `PasswordRecoveryService`) publish notifications through `INotificationPublisher` (see 05.1 §9).
- **Other modules → Mail:** `TaskService` sends reassignment mails as the actor; the TaskFlow Assistant uses mail contacts and drafts (see 05.2 §9).
- **Activity Log:** this module writes no task activity entries. The reminder history (`ReminderHistory`) is the audit record for reminders. Denied requests are logged by `DeniedActionAuditHandler`.

10. Configuration

Shared settings. Child-specific keys are in each child document.

Key (`appsettings.json`)	Purpose	Default
`Storage:Root`	Storage root for the database and attachments	`""` (the platform default from `StorageOptions.DefaultRoot`)
`Application:Timezone`	Fallback application timezone for reminder days and mail dates	`Asia/Kolkata`
`ConnectionStrings:Default`	SQLite database	`<secret>`

Runtime settings held in the `ApplicationSettings` table, which are edited in the Settings module:

Key	Purpose
`Notification.MailEnabled`	Also delivers reminders and comment alerts to Mail
`Application.Timezone`	Application timezone

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11. Testing

Module-wide and cross-module tests. Child-specific test files are listed in each child document.

Test Type	File / Location	Covers
Unit	`tests/TaskFlow.UnitTests/Monitoring/TaskManagement/Discussion/TaskCommentActivityNotificationTests.cs`	Comment notifications and mail (cross-module, both children)

> ⚠️ To be confirmed: the "Covers" column is inferred from file names only, because the test sources were not available.

12. Known Limitations / Future Enhancements

- Reminders are only in-app plus internal Mail. Outbound email (SMTP) was removed by design in the `AddInternalMail` migration.

Gaps found (module-wide)

1. **The TDD `email_delivery_logs`, §11.3 SMTP channel and the `/api/v1/settings/email*` endpoints no longer exist.** The `EmailDeliveryLogs` table was dropped by `20261017090431_AddInternalMail`. BRD Screen 14 is marked SUPERSEDED, so its FRs 1–4, 6 and 8 (SMTP config, secure credentials, test email, delivery attempt logging, config errors) are intentionally not implemented.
2. **Permission audit paths are stale.** `permission-inventory.csv` and `dashboard-feature-inventory.csv` reference old paths (`features/mail/...`, `features/notifications/...`, `core/layout/...`, `TaskFlow.Infrastructure/...`). The code now lives in `modules/communication/...` and `layouts/...`. The keys and behaviour they record are still accurate.

Child-specific gaps are listed in each child document.

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